



SCHOOL OF COMPUTER TECHNOLOGY

AASD 4008 Big Data Tools and Techniques

Topic 1 – Introduction to Big Data



Topic Covered

This section will mainly cover the introduction to big data and discuss the application of big data and the five stages of big data analysis. We will also work with Apache Hadoop and discuss the five pillars of Hadoop. We will use Hive and PIG for relational data and MongoDB for Semi-Structured data.

- Introduction to Big Data
- Evolution of Data
- 5Vs of Big Data
- Types of Data
- Sector wise adaption of Big Data Application
- Applications of Big Data analysis
- Stages in Big Data Analysis
- Types of Big Data Analytics

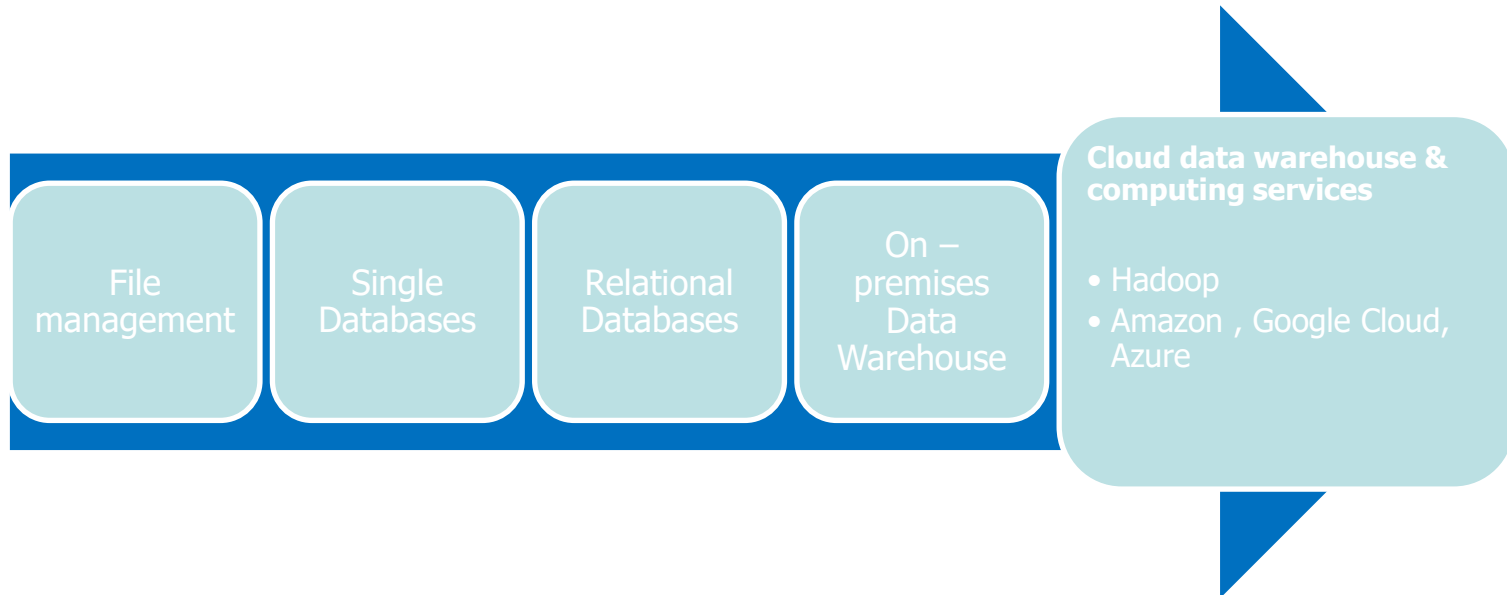
Introduction to Big Data

Big data is a term for *collecting data sets so large and complex that it becomes difficult to process using on-hand database system tools or traditional data process application*

- There is just too much data to process with the traditional system, and it is impossible to maintain, read, and update.
- Big Data is data when it's too complex to be processed by standard computers.

Evolution of Data

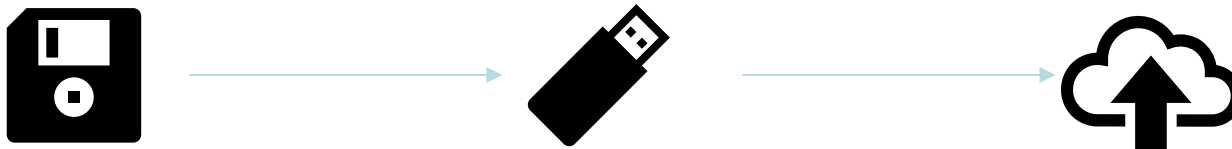
Traditional data has grown into big data in the last decade from a combination of business requirements and technological innovations. Many organizations have used big data either directly or indirectly to their successful operations.



Evolution of Data – continued

Evolution of technology

- We used to have analog telephone sets, now smart phones
- Desktop used to process megabytes of data now; they can process terabytes
- Cars are now smart cars
 - How much data being generated and utilized
 - How many sensors and images must a car process on each turn
 - When to take a turn, when to stop, when to merge, and when to exit
 - It has a sensor for all details and obstacles, and it also sends data home

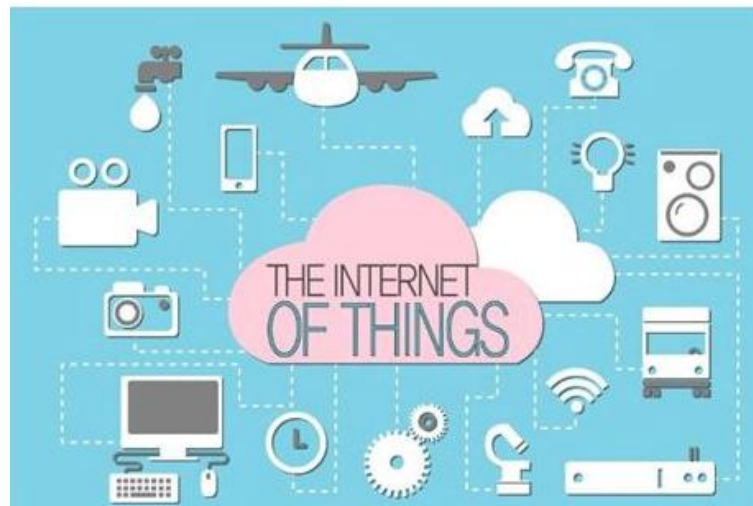


Evolution of Data – continued

Internet of Things(IOT)

IOT devices are connected devices, sensors and algorithms. They all operate in ways that involve a massive amount of data collection, transformation, and processing. The internet of things doesn't just connect physical objects like devices or sensors- it include places like classrooms and bus stops and even intangible objects like a store's payment process

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Samsung.com : Last updated Nov 23, 2020 : https://www.samsung.com/hk_en/support/home-appliances/what-is-internet-of-things-iot/



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Evolution of Data – continued

Data evolved to big data in traditional systems

Data is evolved in traditional / relational systems, there used to be main frame servers that could take care of the entire business, and it may still be true but more and more traditional systems are now looking into or already moved on to big data systems

Retail data shopping : Amazon, Walmart, and many other realtors (online or brick and motor) now have a massive amount of structured and unstructured data that cannot be maintained in traditional systems.

Banking and finances : Digitization in the finance sector has enabled technology that requires big data to penetrate and transform the playing field for institutions to compete for better, fast, and reliable services.

Insurance : Insurance brokers and underwrites are analyzing trends and patterns to dictate policy, assess risk, and identity fraud.

Transportation: Transportation providers who want to optimize operations, cut costs, increase revenues, and better customer satisfaction are using big data tools and technologies. For example, a better understanding of how many customers have used which route and how often will enable transportation providers to deploy staff and vehicles on busy routes.

Government: Statistics Canada is integrating big, structured data, such as administrative data, to understand the outcome of immigration better. They are also using big data to understand labor market outcomes of post-secondary grads. **Another example** of big data application is to use satellite data to provide timely information on crop yields.

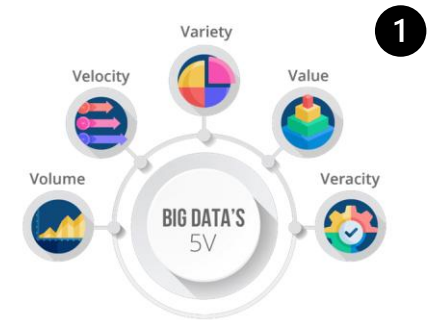
Healthcare : They used big data for predictive analysis to guide options for cancer screening.



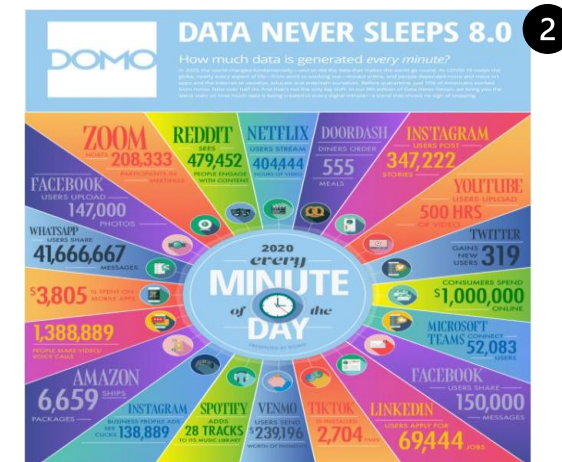
5Vs of big data

Who ever came up with the terms 5Vs of bigdata has made the definition of big data quite clear !

Volume	The amount of data is in exabytes, petabytes or more. Big data technology giants like Amazon, Shopify, and other e-commerce platforms get real-time data, laying between terabytes and zettabytes every second from millions of customers across the globe.
Variety	Different types of data are being generated from various sources. Tables/JSON/MP3/Logs/Comments/Emails/Click Stream / Likes/ Page visits Structured – Semi Structured – un Structured.
Velocity	Speed of accumulation, the transmission, the rate the data being generated is categorized under velocity. 500 million tweets sent per day.
Value	Data useless until and unless its being transform into something valuable. Value is a mechanism to bring the correct meaning out of the data, steps are to Mine Useful Data -> analysis – >extract Value Netflix records your browsing pattern, pauses, rewind, and forwards to create a profile and provide valuable recommendations.
Veracity	Uncertainty and inconsistencies in the data Veracity is to deal with the “truthiness” or “messiness” of the data.



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Medium.com : <https://medium.com/analytics-vidhya/5vs-of-big-data-to-a-non-it-person-c498f8b56c86>

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Domo.com : Infographic : Data Never Sleeps 8.0 - <https://www.domo.com/learn/infographic/data-never-sleeps-8>

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5Vs of big data- Explain

Value	Velocity	Variety	Veracity	Value
Big data refers to the vast volume of data being generated from various sources such as social media, sensors, and online transactions. It involves dealing with extremely large datasets that often exceed the capabilities of traditional data processing systems.	Velocity represents the speed at which data is generated, collected, and processed. Big data often arrives in real-time or near real-time, requiring efficient processing and analysis techniques to extract meaningful insights and respond quickly to changing conditions.	Variety refers to the diverse types and formats of data that are part of big data. It includes structured data (e.g., relational databases), semi-structured data (e.g., XML, JSON), and unstructured data (e.g., text documents, images, videos). Dealing with the variety of data requires flexible and scalable processing methods.	Veracity refers to the quality, reliability, and trustworthiness of the data. Big data often contains noisy, incomplete, or conflicting information. Ensuring data quality and addressing data inconsistencies is crucial to derive accurate insights and make reliable decisions.	Value represents the potential insights, benefits, and value that can be derived from analyzing big data. Extracting actionable insights from large datasets can lead to improved business strategies, enhanced decision-making, cost savings, operational efficiency, and innovation.

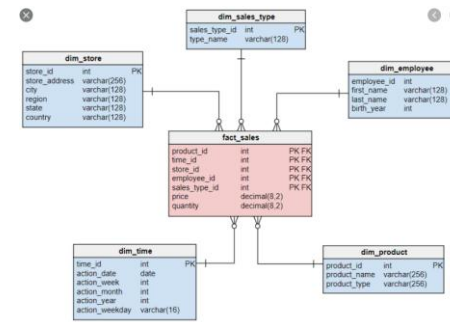
Types of Data

There are 3 Main type of data – classified via their structure and adaptability

Structured : Structured data comprises of numbers or labels stored in a structured framework of columns and rows relating to pre-set parameters, follow a rigid schema. They are neatly arranged and ready for query.

Following are the example of structured data

- Databases
- Excel Sheets
- CSV files



Types of Data – Continued

Semi - Structured : Semi-structured data is loosely organized into categories using metadata tags. It has no pre-defined format or organization ,making analysis and adaptability a bit more difficult than structured data.

- Emails
- Tweets
- XML
- JSON

```
{
  "custkey": "450002",
  "useragent": {
    "devicetype": "pc",
    "experience": "browser",
    "platform": "windows"
  },
  "pagetype": "home",
  "productline": "television",
  "customerprofile": {
    "age": 20,
    "gender": "male",
    "customerinterests": [
      "movies",
      "fashion",
      "music"
    ]
  }
}
```



Following are the examples

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Sector wise adaptation of Big Data applications



Healthcare	Self-awareness records for OHIP. Blood works results.
Telecom	Analysis of all call records and identify fraud detection. It also uses comments data for better customer support.
Insurance	Use big data analysis for risk assessments, fraud detection.
Finance	Differentiate between normal and abnormal transaction? How many times your card is blocked or rejected a transaction
Automobile	Sensors in engines...schedule a maintenance !!!
Retail	Expedia has different officers for different customers. Starbucks keeps a history of customer's pattern choices. When you go to a new store, the point of sales system identifies the customer.



Sector wise adaptation of Big Data applications - HealthCare



The adoption of big data **in healthcare** has been transformative, offering new opportunities for improving patient care, research, and operational efficiency.

Clinical Decision Support: Big data analytics enables healthcare providers to **analyze large volumes of patient data, including electronic health records (EHRs), medical images, genetic data, and wearable device data.** This helps in identifying **patterns, predicting disease risks,** and providing **personalized treatment recommendations,** enhancing clinical decision-making and patient outcomes.

Population Health Management: Big data allows healthcare organizations **to analyze population-level data to identify trends, patterns, and risk factors.** This helps in proactively managing the health of specific populations, identifying high-risk individuals, and designing targeted interventions to improve overall population health and reduce healthcare costs.

Drug Discovery and Development: Big data analytics plays a vital role in **drug discovery and development.** It enables **researchers to analyze vast amounts of genomic data,** clinical trial data, scientific literature, and real-world evidence to identify potential drug targets, predict drug responses, and accelerate the discovery of new therapies.



Sector wise adaptation of Big Data applications - Insurance



Risk Assessment and Underwriting: Big data analytics plays a significant role in insurance by enhancing risk assessment and underwriting processes. Insurance companies can leverage diverse data sources, including customer demographics, historical claims data, credit scores, social media activity, and external data like weather patterns and economic indicators. By analyzing this vast amount of data, insurers can gain insights into individual risk profiles, accurately assess risks, and make more informed underwriting decisions. Big data analytics enables insurers to price policies more accurately, identify potential fraud, and offer personalized insurance products to customers.

Claims Management and Fraud Detection: Big data analytics is also valuable in claims management and fraud detection in the insurance industry. By analyzing claims data, including historical claims records, policy information, medical data, and external data sources, insurers can detect patterns and anomalies that may indicate fraudulent activity. Big data analytics enables real-time monitoring of claims, identification of suspicious claims patterns, and automation of fraud detection processes. This helps insurance companies streamline claims management, reduce fraudulent claims, and ensure fair compensation to legitimate claimants.



Sector wise adaptation of Big Data applications - AutoMobile



Connected Car Analytics: Big data plays a crucial role in the automotive industry through connected car analytics. Modern vehicles are equipped with sensors and telematics systems that generate a vast amount of data on vehicle performance, driver behavior, maintenance needs, and navigation patterns. By analyzing this data, automakers and service providers can gain insights into vehicle health, identify potential issues, optimize maintenance schedules, and offer predictive maintenance services. Big data analytics also enables real-time monitoring of vehicle performance, allowing for proactive interventions and ensuring a safer and more efficient driving experience.

Improving Product Development: Big data analytics is transforming product development processes in the automobile industry. Automotive companies collect data from various sources, including customer feedback, social media sentiment, and vehicle testing data, to gain insights into customer preferences, market trends, and product performance. By analyzing this data, automakers can make data-driven decisions on design improvements, features, and future product strategies. Big data analytics also helps in conducting simulations, virtual testing, and predictive modeling, enabling faster innovation, reducing development cycles, and bringing new products to market more efficiently.



Sector wise adaptation of Big Data applications – Financial Applications

Algorithmic Trading and Market Analysis: Big data analytics is extensively used in algorithmic trading and market analysis. Financial institutions leverage vast amounts of market data, including historical trading data, real-time market feeds, news articles, and social media sentiment, to identify trends, patterns, and trading opportunities. By analyzing this data, financial firms can develop sophisticated trading algorithms that automate trading decisions, optimize trade execution, and capitalize on market inefficiencies. Big data analytics also helps in conducting market research, predicting market movements, and enhancing investment strategies.

Personalized Financial Services: Big data analytics allows financial institutions to offer personalized financial services and targeted marketing campaigns to their customers. By analyzing customer data, including transaction history, spending patterns, investment preferences, and demographic information, financial institutions can create customized offers, recommend suitable products, and provide personalized financial advice. Big data analytics enables a deeper understanding of customer needs and behaviors, leading to improved customer satisfaction, increased cross-selling opportunities, and enhanced customer loyalty.



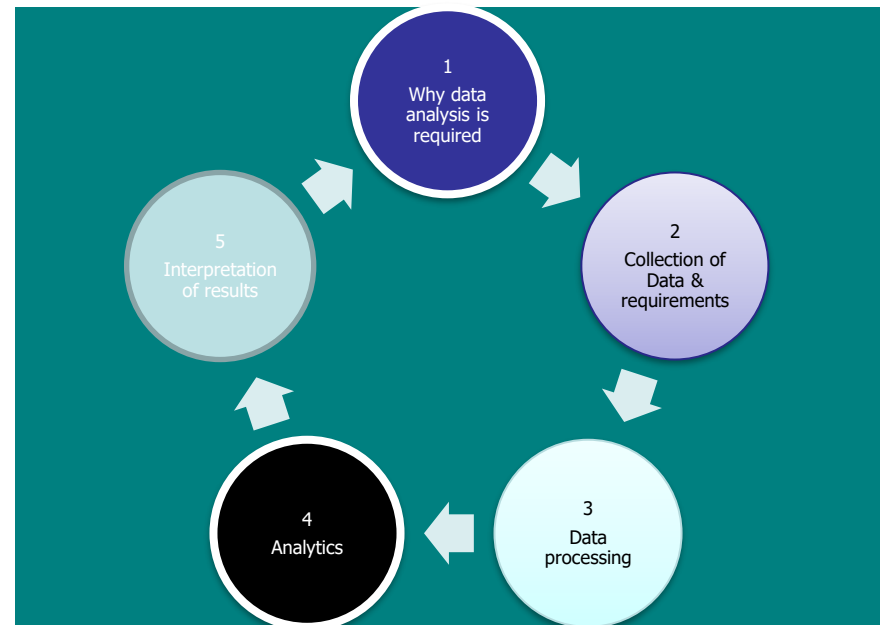
Applications of big data analysis



NetFlix	Netflix uses big data analytics to analyze user viewing patterns, preferences, and ratings to personalize recommendations and improve content suggestions. Their algorithms process massive amounts of data to deliver a personalized user experience.
Uber	Uber uses big data to optimize its ride-sharing platform. They analyze real-time data on driver locations, passenger demand, and traffic conditions to match drivers with riders efficiently and estimate arrival times accurately.
Facebook	Facebook utilizes big data analytics to personalize users' news feeds, target advertising, and understand user behavior. They analyze user interactions, interests, and demographic information to deliver relevant content and improve the user experience.
Google	Google uses big data extensively for its search engine algorithms, advertising platforms, and personalized services. They analyze user search queries, browsing behavior, and location data to deliver relevant search results and targeted advertisements.
Tesla	Tesla uses big data analytics to improve its electric vehicles' performance and reliability. They collect and analyze data from sensors in their vehicles to identify patterns, optimize battery life, and enhance autonomous driving capabilities.
AirBnb	Airbnb utilizes big data to enhance its platform and improve the user experience. They analyze data on property listings, user reviews, and booking patterns to provide personalized recommendations and optimize pricing strategies.
LinkedIn	LinkedIn uses big data analytics to provide relevant job recommendations, networking suggestions, and personalized content to its users. They analyze user profiles, connections, and job market data to offer tailored recommendations and insights.
Spotify	Spotify leverages big data to curate personalized playlists and recommendations for its users. They analyze user listening habits, preferences, and music data to deliver personalized music recommendations and discover new songs and artists.
Twitter	Twitter utilizes big data analytics to understand user engagement, sentiment analysis, and trending topics. They analyze millions of tweets to identify popular conversations, track user sentiment, and deliver personalized content to users.
American Express	American Express utilizes big data analytics to detect fraudulent transactions and provide personalized offers to its customers. They analyze transaction data, spending patterns, and customer behavior to identify suspicious activity and deliver tailored rewards and offers.

Stages in Big Data Analysis

1-Identify problem	What is the problem we need to solve? Why you need the data analysis.
2-Collecting data & requirements	What kind of data is required, where do you get it
3-Preprocessing data	Cleaning of data. Some processing, ETL (Extract, Transform and Load)
4-Perform analytics over data	Machine learning. Deep learning
5-Visualizing data	Dashboard....



Type of Big Data Analytics

Data Analysis is a process involving collecting , manipulating , and examining data for getting deep insight

Data Analytics is taking the analyzed data and working on it a meaningful way

While there is no shortage of data out there ,the ability to access and act on it whenever, wherever and, however, is called big data analytics !



Type of Big Data Analytics Continued - Diagnostic



Descriptive	What is happening now and based on incoming data and historical data ? Google analytics tool is the best example for descriptive analysis. A business gets result from the web server through the tool, which helps understand what happened in the past and validates if a promotional campaign was successful or not based on basic parameters like page view.
Predictive	What might happen in the future? Southwest airlines analyzes sensor data on their planes to identify patterns that show a potential malfunction, thus allowing the airlines to the repairs before its schedule.
Prescriptive	What action should be taken? Google's self-driving car is a perfect example of prescriptive analytics. It analyzes the environment and decides the direction to take based on data.
Diagnostic	Why did it happen? For a social media marketing campaign, you can use diagnostic analytics to assess the number of posts, mentions, followers, fans, page views and reviews, pings, etc....and analyze the failure and success rate of the campaign at a fundamental level



Type of Big Data Analytics Continued- Descriptive



Descriptive data analytics is a branch of data analytics that focuses on summarizing and interpreting historical data to gain insights and understand patterns. It involves analyzing and visualizing data to uncover trends, relationships, and key characteristics. Here are two main points on descriptive data analytics with examples:

Data Visualization: Descriptive data analytics often involves the use of data visualization techniques to represent complex data in a visual and easily understandable format. This allows analysts and decision-makers to gain insights quickly and identify patterns or anomalies. For example, a retail company may use data visualization to create interactive dashboards that display sales trends, customer demographics, and product performance, enabling them to identify the best-selling products or understand the buying behavior of different customer segments.

Summary Statistics: Descriptive data analytics also includes the use of summary statistics to summarize and describe the characteristics of a dataset. These statistics provide a concise overview of the data, such as measures of central tendency (e.g., mean, median) and measures of variability (e.g., standard deviation, range). For instance, a financial institution may analyze the monthly credit card transaction data to calculate the average transaction amount, identify the most frequent transaction categories, and determine the standard deviation to assess the variability in spending patterns.

In both examples, descriptive data analytics is used to summarize and interpret historical data, enabling businesses to gain valuable insights and make informed decisions. By visualizing data and deriving summary statistics, organizations can understand trends, identify patterns, and extract actionable information from their datasets.



Type of Big Data Analytics Continued- Predictive



Predictive data analysis, also known as predictive analytics, is a branch of data analytics that utilizes historical data and statistical modeling techniques to make predictions or forecasts about future events or outcomes. Here are two points on predictive data analysis in big data:

Predictive Modeling: Predictive data analysis involves the creation and deployment of predictive models that utilize algorithms to analyze large volumes of historical data and make predictions about future events or behaviors. These models can identify patterns, relationships, and trends in the data and use them to forecast outcomes. For example, in the healthcare industry, predictive data analysis can be used to develop models that predict the likelihood of patient readmission, or the progression of a disease based on patient data and medical records.

Decision Support and Optimization: Predictive data analysis enables organizations to make data-driven decisions and optimize their operations. By analyzing big data, organizations can develop predictive models to optimize various aspects of their business, such as inventory management, supply chain optimization, and resource allocation. For instance, in the retail industry, predictive data analysis can be used to forecast customer demand, optimize pricing strategies, and improve inventory management by identifying the right products to stock and when to restock them.

Predictive data analysis in big data empowers organizations to anticipate future outcomes, optimize processes, and make informed decisions. By leveraging historical data and predictive models, organizations can gain a competitive advantage by identifying opportunities, mitigating risks, and improving overall operational efficiency.



Type of Big Data Analytics Continued- Prescriptive



Decision Optimization: Prescriptive data analysis combines historical data, predictive models, and optimization algorithms to provide recommendations on the best course of action to achieve specific goals or objectives. It considers various constraints and objectives to identify the optimal decision or action to take. For example, in logistics and transportation, prescriptive data analysis can optimize delivery routes, considering factors such as traffic conditions, delivery schedules, and vehicle capacities, to minimize costs and improve efficiency.

Real-Time Decision Support: Prescriptive data analysis can be applied in real-time scenarios, leveraging big data and advanced analytics techniques to provide real-time decision support. By continuously analyzing streaming data and combining it with historical data and predefined business rules, organizations can receive recommendations or alerts for immediate action. For instance, in fraud detection, prescriptive data analysis can analyze real-time transaction data, detect patterns of fraudulent activity, and trigger alerts or automated actions to prevent or mitigate potential fraud in real-time.



Type of Big Data Analytics Continued- Diagnostic



Diagnostic analysis in big data analytics involves examining historical data to understand the reasons behind specific events or outcomes. It focuses on identifying the root causes and factors contributing to certain patterns or behaviors observed in the data. Here are two points on diagnostic analysis in big data analytics:

Root Cause Analysis: Diagnostic analysis helps organizations identify the root causes of certain events or outcomes by analyzing big data. It involves exploring relationships, correlations, and dependencies within the data to uncover the underlying factors driving the observed patterns. For example, in customer churn analysis, diagnostic analysis can be used to investigate the factors that contribute to customer attrition by examining historical customer data, such as purchase history, customer interactions, and demographic information. This analysis can reveal insights into the specific reasons behind customer churn, such as poor customer service or pricing issues.

Anomaly Detection: Diagnostic analysis can also be used to identify anomalies or deviations from expected behavior within big data. By comparing current data to historical data or established norms, organizations can detect unusual patterns or outliers. This helps in understanding the factors contributing to the anomalies and investigating potential issues or opportunities. For instance, in cybersecurity, diagnostic analysis can detect anomalous network behavior or unauthorized access attempts by analyzing network traffic data, system logs, and user behavior, allowing organizations to identify security breaches or vulnerabilities.

Diagnostic analysis in big data analytics helps organizations gain a deeper understanding of the underlying factors and reasons behind observed patterns or events. By analyzing historical data and identifying root causes or anomalies, organizations can make informed decisions, address issues proactively, and optimize their processes for improved performance.



Cloud Computing (Azure – AWS – GCP)



Cloud computing is the on-demand delivery of IT resources over the Internet with pay-as-you-go pricing. Instead of buying, owning, and maintaining physical data centers and servers, you can access technology services, such as computing power, storage, and databases, on an as-needed basis from a cloud provider like Azure , AWS and GCP etc



Cloud Computing Implementations



Organizations of every type, size, and industry are using the cloud for a wide variety of use cases, such as data backup, disaster recovery, email, virtual desktops, software development and testing, big data analytics, and customer-facing web applications.



Cloud Computing Benefits



Agility

The cloud gives you easy access to a broad range of technologies so that you can innovate faster and build nearly anything that you can imagine. You can quickly spin up resources as you need them—from infrastructure services, such as compute, storage, and databases, to Internet of Things, machine learning, data lakes and analytics, and much more.

You can deploy technology services in a matter of minutes, and get from idea to implementation several orders of magnitude faster than before. This gives you the freedom to experiment, test new ideas to differentiate customer experiences, and transform your business.



Cloud Computing Benefits



Elasticity

With cloud computing, you don't have to over-provision resources up front to handle peak levels of business activity in the future. Instead, you provision the amount of resources that you actually need. You can scale these resources up or down to instantly grow and shrink capacity as your business needs change.



Cloud Computing Benefits



Deploy globally in minutes

With the cloud, you can expand to new geographic regions and deploy globally in minutes. For example, AWS has infrastructure all over the world, so you can deploy your application in multiple physical locations with just a few clicks. Putting applications in closer proximity to end users reduces latency and improves their experience.



Cloud Computing Benefits



Cost savings

The cloud allows you to trade fixed expenses (such as data centers and physical servers) for variable expenses, and only pay for IT as you consume it. Plus, the variable expenses are much lower than what you would pay to do it yourself because of the economies of scale.



Cloud Computing Types

The three main types of cloud computing . Each type of cloud computing provides different levels of control, flexibility, and management so that you can select the right set of services for your needs.



Infrastructure as a Service (IaaS)

IaaS contains the basic building blocks for cloud IT. It typically provides access to networking features, computers (virtual or on dedicated hardware), and data storage space. IaaS gives you the highest level of flexibility and management control over your IT resources. It is most similar to the existing IT resources with which many IT departments and developers are familiar.



Platform as a Service (PaaS)

PaaS removes the need for you to manage underlying infrastructure (usually hardware and operating systems), and allows you to focus on the deployment and management of your applications. This helps you be more efficient as you don't need to worry about resource procurement, capacity planning, software maintenance, patching, or any of the other undifferentiated heavy lifting involved in running your application.

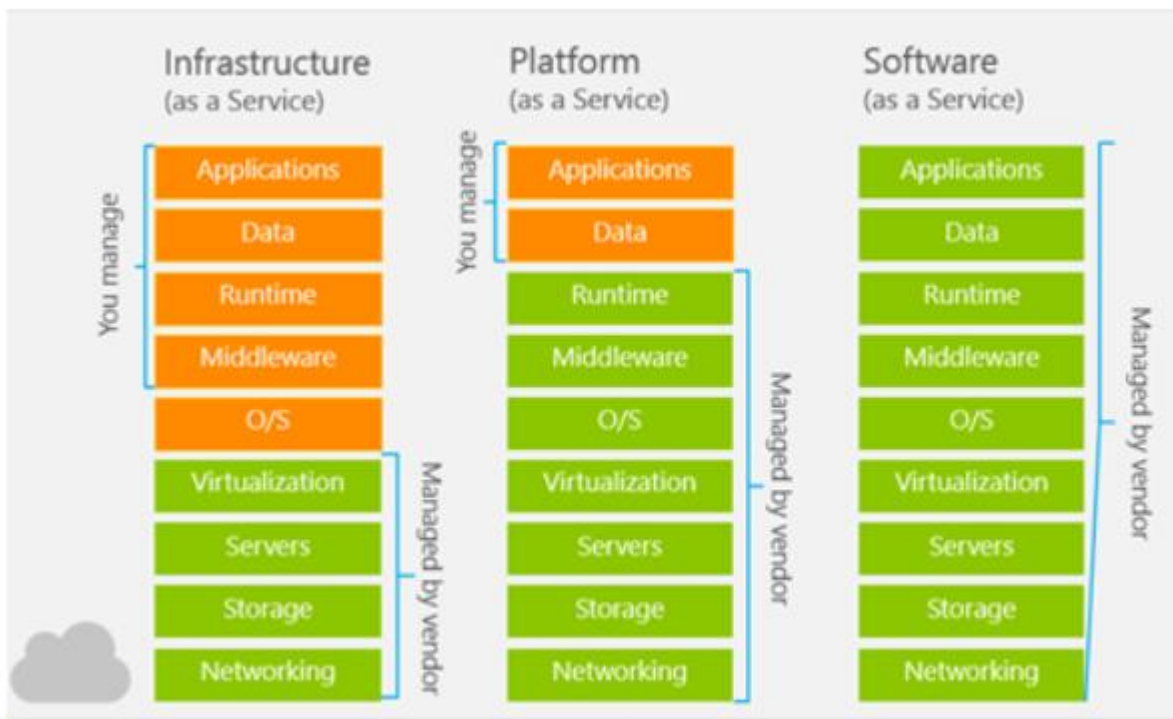


Software as a Service (SaaS)

SaaS provides you with a complete product that is run and managed by the service provider. In most cases, people referring to SaaS are referring to end-user applications (such as web-based email). With a SaaS offering, you don't have to think about how the service is maintained or how the underlying infrastructure is managed. You only need to think about how you will use that particular software.



Cloud Computing Types



Credits and Citations



- Some material and image are taken from Edureka website and their YouTube tutorials.
<https://www.edureka.co/>
- Some material and image are taken from simplilearn website and their YouTube tutorials.
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- <https://www.cselectricalandelectronics.com/big-data-evolution-of-big-data-benefits-of-big-data-opportunities/>
- zdnet.com (<https://www.zdnet.com/article/ten-examples-of-iot-and-big-data-working-well-together>)
- Big Data Framework.org (<https://www.bigdataframework.org/business-drivers-for-big-data>)
- Apache Pig <https://pig.apache.org/>